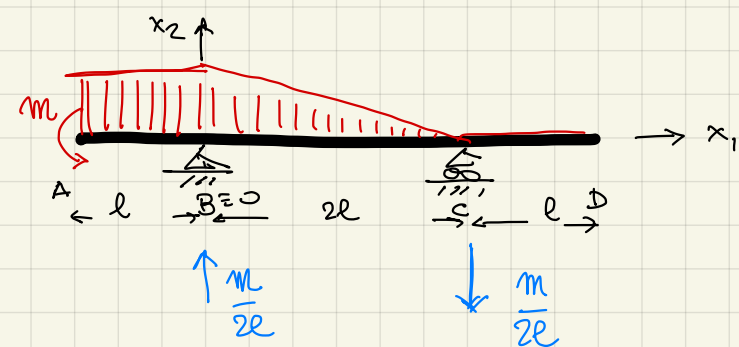


Exercise 1



ISO

$$u(B) = w(B) = 0$$

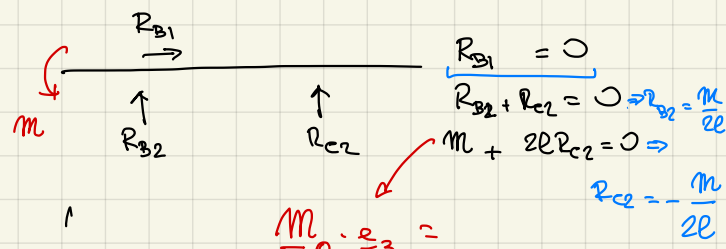
$$w(C) = 0$$

$$\underline{x}(B) = \underline{0}, \quad \underline{x}(C) - \underline{e}_2 = 0$$

[Example of a mechanism]



FBD



$$T(x) = \begin{cases} 0 & x \in (-l, 0) \\ -\frac{m}{2l} & x \in (0, 2l) \\ 0 & x \in (2l, 3l) \end{cases}$$

$$M' + T + m = 0 \Rightarrow M' = -T \dots \text{integrate}$$

$$M'' = f_2 \equiv 0$$

$$M(x) = \begin{cases} -m & x \in (-l, 0) \\ m(\frac{x}{2l} - 1) & x \in (0, 2l) \\ 0 & x \in (2l, 3l) \end{cases}$$

$$\varphi_B = ? = w'(0)$$

$$w''(x) = \frac{M(x)}{EI} \quad \text{piecewise}$$

Integrate in $x \in (0, 2l)$ with BC $w(0) = 0, w(2l) = 0$

$$w''(x) = \frac{m}{EI} \left(\frac{x}{2l} - 1 \right) \rightarrow w'(x) = w'(0) + \frac{m}{EI} \frac{x^2}{4l} - \frac{m}{EI} x$$

$$w(x) = \cancel{w(0)} + w'(0)x + \frac{m}{EI} \frac{x^3}{12} - \frac{m}{EI} \frac{x^2}{2} \quad (*)$$

$$w(2l) = 0 = w'(0)2l + \frac{m}{EI} \frac{8}{12} l^3 - \frac{m}{EI} \frac{4l^2}{2}$$

$$w'(0) = \frac{m}{EI} l \left(1 - \frac{1}{3} \right) = \frac{m}{3EI} 2l = \varphi_B$$

Check: plug $w'(0)$ in (*) and check that, indeed, $w(2l) = 0$

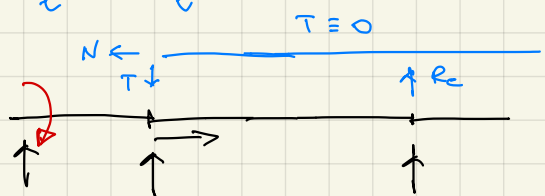
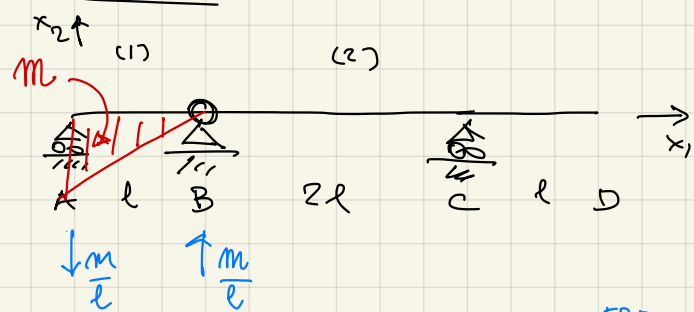
Other exercises:

$$w_A, \varphi_A$$

$$\varphi_C$$

$$\varphi_D, w_D$$

Exercise 2



FBD (partial: $(l, 4l)$, using that $M(l) = 0$!)

FBD (global: $(0, 4l)$, 4 unknowns are too many for 3 equations!)

$$\begin{aligned} w_A &= 0 \\ w_B &= 0 \\ u_B &= 0 \\ w_C &= 0 \end{aligned}$$

$$\begin{aligned} w_B^+ &= w_B^- \\ u_B^+ &= u_B^- \end{aligned}$$

$$[\varphi_B^+ = \varphi_B^-]$$

Load: Ext. couple M_{E3}

applied at A

Compute: $\varphi_B^+ =$, $\varphi_B^- =$

$$M(x) = \begin{cases} m - \frac{m}{l}x & x \in (0, l) \\ 0 & x \in (l, 3l) \\ 0 & x \in (3l, 4l) \end{cases}$$

$$w''(x) = \frac{f^{\wedge}(x)}{EI} \text{ piecewise}$$

$x \in (l, 4l)$: $M(x) \equiv 0 \Rightarrow w(x) \text{ affine} \equiv 0$ because $w(l) = w(3l) = 0 \Rightarrow w \equiv 0$ in $(l, 3l)$

hence $w(3l) = w'(3l) = 0 \Rightarrow w \equiv 0$ in $(3l, 4l)$

$$\Rightarrow \boxed{\varphi_B^+ = 0}$$

$x \in (0, l)$

$$w''(x) = \frac{m}{EI} \left(1 - \frac{x}{l} \right) \Rightarrow w'(x) = w'(0) + \frac{m}{EI} \left(x - \frac{x^2}{2l} \right)$$

$$w(x) = \underbrace{w(0)}_{=0} + w'(0)x + \frac{m}{EI} \left(\frac{x^2}{2} - \frac{x^3}{6l} \right)$$

$$w(l) = 0 = w'(0)l + \frac{m}{EI} \left(\frac{l^2}{2} - \frac{l^3}{6l} \right) \Rightarrow w'(0) = \frac{m}{EI} \left(\frac{1}{6} - \frac{1}{2} \right)$$

$$\frac{1-3}{6} = -\frac{2}{6} = -\frac{1}{3}$$

$$[\varphi_A = -\frac{m}{3EI}]$$

$$w(x) = -\frac{m}{3EI}x + \frac{m}{EI} \left(\frac{x^2}{2} - \frac{x^3}{6l} \right)$$

$$w'(x) = -\frac{m}{3EI} + \frac{m}{EI} \left(x - \frac{x^2}{2l} \right)$$

$$-\frac{1}{3} + 1 - \frac{1}{2} = \frac{-2+6-3}{6} = \frac{1}{6}$$

$$w'(l) = \varphi_B^- = -\frac{m}{3EI} + \frac{m}{EI} \left(l - \frac{l^2}{2l} \right) = -\frac{m}{3EI} + \frac{m}{EI} \left(1 - \frac{1}{2} \right) =$$

$$= \boxed{\frac{m}{6EI} = \varphi_B^-}$$